

COS 111 Work Scheme

Weeks	Topics
1	Core concepts of computing. Identification of problems. Types of problems (routine problems and non-routine problems).
2	Problem-solving. Methods of solving computing problems. Solution techniques of solving problems. Divide and Conquer, Top-Down Approach
3	Algorithms: • Role of Algorithms in problem solving process • Definition and concepts of algorithms • Properties of an algorithm
3	i. Flowcharts: • Definition • Flowchart Symbols • Flowchart Developments • Working with examples ii. Pseudo-codes: • Definition • Examples
4	Working with examples on algorithm writing • Sequential approach • Scenario-based problems
5	Working with examples on algorithm writing • Conditional approach • Scenario-based problems
6	Working with examples on algorithm writing • Iterative approach • Scenario-based problems
	1 st Test
7	Coding • Introduction to programming concepts
8 & 9	Introduction to Python programming language
10 - 11	Practical Programming classes with examples
12	2 nd Test and Revision
13	Examination

Tutorial Questions

Question 1

- (a) Write an algorithm to sum a series of (positive) integers entered by the user, excluding all numbers that are greater than 100. **(10marks)**
- (b) Draw the flowchart for the algorithm **(5marks)**

Question 2

(a) Classify algorithm into three (3) based on control structure and illustrate each class with a detailed example.

(3 mark for classification and highlight, 2 marks for each example; Total: 9marks)

(b) Explain the four (4) basic steps involved in problem solving? **(6 marks)**

Question 3

(a) Discuss the three (3) classification of programming languages base on their features

(3 marks each, total 9marks)

(b) Define flowchart **(1.5mark)** and explain the use of any three (3) of the components of the building block **(1.5 marks each: 4.5marks)**

Question 4

a) Write an algorithm to find the average of N given numbers **(5marks)**

b) Draw the flowchart for the algorithm in (a). **(5 marks)**

c) Write an algorithm to find the root of a quadratic equation: $ax^2 + bx + c = 0$ **(5marks)**

Hint: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $d = \text{sqrt}(b^2 - 4ac)$

Question 5

a) Discuss the characteristics of algorithm. **(5marks)**

(b) Write a Python program in which the user enters either 'A', 'B', or 'C'. If 'A' is entered, the program should display the word 'Apple'; if 'B' is entered, it displays 'Banana'; and if 'C' is entered, it displays 'Coconut'. **(10marks)**

Question 6

Given the python code snippet;

```
if my_var % 2:
    if my_var**3 != 27:
        my_var = my_var + 4
    else:
        my_var /= 1.5
else:
    if my_var <= 10:
        my_var *= 2
    else:
        my_var -= 2
print(my_var)
```

(a) Explain each line of the code and what it does **(5marks)**

(b) Supply four values of my_var so that each of the four assignment statements will be executed: each value should cause one assignment statement to be executed. **(8marks)**

(b) Explain how nested-if works. **(2marks)**

Question 7

- a) Write an algorithm that prompts the user for the age of an individual and display the category the person belongs.

Hint on age groups: Age 0-5: Toddler, Children, Age 6-11, Teenager: Age 12- 19, Adult: Age 20 and above (7.5marks)

- b) Draw the flowchart for the algorithm in (a) above (5marks)
c) Define variable and enumerate the rules guiding naming of variable in python. (2.5marks)

Question 8

- (a) Evaluate the Boolean expressions below for $n = 10$ and $k = 20$.

- i. $(n > 10) \text{ and } (k == 20)$
- ii. $(n > 10) \text{ or } (k == 20)$
- iii. $\text{not } ((n > 10) \text{ and } (k == 20))$
- iv. $\text{not } (n > 10) \text{ and not}(k == 20)$
- v. $(n > 10) \text{ or } (k == 10 \text{ or } k != 5)$ (2marks each: 10marks)

- (b) Define the followings and give example:

- i. Assignment statement
- ii. An expression
- iii. Variable
- iv. A program
- v. An operator (1mark each: 5marks)

Question 9

- d) List and explain six fundamental features of object-oriented programming.

(Listing: 1mark, Explanation: 1.5marks each, total 10marks)

- e) Highlight five differences between structured programming and object-oriented programming. (5 marks)

Question 10

- (a) Write a Python program that prompts the user for two integer values and displays the result of the first number divided by the second, with exactly two decimal places displayed.

(7.5marks)

- (b) Draw a flowchart for the problem in (a) (7.5marks)

Question 11

- (ai) If you execute the three lines of code given below, what will be printed? (1mark)

```
my_int = 5
my_int = my_int + 3
print (my_int)
```

- (ai) Explain your answer using the rules of assignment for each line (3marks)

- (bi) Rewrite `my_int = my_int + 3` using the `+=` symbol. (2marks)

- (bii) Explain the difference between a statement and an expression. Give an example of both, and explain what is meant by a statement having a *side effect*. (3marks)

- (c) Differentiate between the three (3) different types of translators. (2 marks each: 6marks)

Question 12

If the lengths of the two parallel sides of a trapezoid are X meters and Y meters, respectively, and the height is H meters.

- a) Draw the flowchart to output the area **(8marks)**
- b) Write Python code to output the area. **(7marks)**

Question 13

- (a) Write an algorithm to calculate the simple interest using the formula given. **(5marks)**

Simple interest = $P \times N \times R / 100$ where P = Principal, N = number of years, and R = rate

- (b) Develop the flowchart for the algorithm in (a) above **(5marks)**
- (c) Explain five (5) components of the building block of a flowchart
(Each component with diagram and explanation on usage (1 marks each) – (5marks),

Question 14

- a) List and explain six fundamental features of object-oriented programming.

(Listing: 1mark, Explanation: 1.5marks each, total 10marks)

- b) Highlight five differences between structured programming and object-oriented programming. **(5 marks)**

Question 15

- a) Write a python program where the user can enter three numbers. The program should find the minimum and the maximum of these three numbers. It should then display
“The minimum of these three numbers is” and
“The maximum of these three numbers is ...” and display the appropriate numbers.
(9 marks)

- b) Differentiate between the three (3) different types of translators.
(2 marks each, total 6marks)

Question 16

- a) Write an algorithm to generate and print the first N terms of Fibonacci Series, where N is greater than or equal to 1. **(10marks)**
- b) Draw the flowchart for the algorithm in (a) above **(5marks)**

Question 17

- (a) Discuss the three (3) classification of programming languages base on their features
(3 marks each, total 9marks)
- (b) Name any three (3) general purpose software packages on a personal computer with a brief discussion on their function. **(2marks each, total 6marks)**